MASTER'S COMPREHENSIVE EXAMINATION

Part I **Theory (90 minutes)** **March 24, 2007**

Instructions: Answer all four questions. Open book and open notes.

1. (a) Assume a McDonald's $\frac{1}{4}$ pounder is normally distributed with mean μ and variance .0004. What value of μ is such that the probability that a burger is less than $\frac{1}{4}$ pound is .01?

(b) Let $X \sim N(2,1)$. Find $P\{X > 2 | X > 1\}$.

2. Let $X_1, \ldots, X_n$ be $N(0, \sigma^2)$.

(a) Find a one-dimensional sufficient statistic.

(b) Find a 95% confidence interval for σ based on the sufficient statistic.

3. Let X_1, X_2 be a random sample of size 2 from a normal population with mean μ_x and variance σ_x^2. Let Y_1, Y_2, Y_3 be a random sample of size 3 from a normal population with mean μ_y and variance σ_y^2. Assume the samples are independent and $\sigma_x^2 = \sigma_y^2$.

(a) Find a 95% confidence interval for $\mu_x - \mu_y$.

(b) Find the MSE of the estimator

$$\left(\frac{X_1 + X_2}{2} - \frac{(Y_1 + Y_2 + Y_3)}{3} \right)$$

for $\mu_x - \mu_y$.

4. (a) Let $X_1 \sim B(3, 1/2)$ independent of $X_2 \sim B(2, 1/3)$. Find $P\{X_1 + X_2 = 4\}$.

(b) Let U be uniformly distributed on $(0,1)$. That is,

$$f_U = \begin{cases} 1 & \text{if } 0 < u < 1 \\ 0 & \text{otherwise.} \end{cases}$$

Find the density of $V = -2 \log_e U$.